

Name - Saikat Sheet
Batch - 25SUB5936_Linux System
Programming
ID - 68500

Assignment 2

1. Introduction to network security challenges.

Ans :

Common Challenges

- **Hacking:** Outside users trying to break into your router to see your private data.
- **Viruses:** Harmful software that can spread from your laptop to your phone.
- **Weak Passwords:** Using simple passwords makes it easy for others to guess their way into your Wi-Fi.
- **Outdated Software:** If your devices aren't updated, they may have "holes" that let in threats.

How to Stay Safe

- **Use a Firewall:** This acts like a digital security guard for your router.
- **Strong Passwords:** Use a mix of letters, numbers, and symbols for your Wi-Fi.
- **Updates:** Always click "update" on your phone and laptop to fix security bugs.

2. Basic security measures (Firewalls, VPNs).

Ans:

Firewalls (The Digital Security Guard)

A **Firewall** is a barrier between your home network and the rest of the internet. It acts like a security guard at a gate, checking every piece of data trying to enter or leave.

- **What it does:** it blocks "bad" traffic (like hackers or viruses) while letting "good" traffic (like your Netflix stream) pass through.
- **Where it lives:** Most modern Wi-Fi routers have a firewall built directly into them.

VPNs (The Private Tunnel)

A **VPN** (Virtual Private Network) creates a private, encrypted "tunnel" for your data to travel through.

- **What it does:** It hides your true location and scrambles your information so that nobody—not even your internet provider—can see what you are doing or steal your passwords.
- **When to use it:** It is especially important when using your **laptop** or **mobile phone** on public Wi-Fi (like at a coffee shop).

